

Assignment # 2

Step 1: Problem Statement

Create a program to read from a file to create a Polymorphic Array with the values. After creating the "Actors" array, iterate through values to display Name, Type & Moto in a formatted table.

A "Movie" Object will then be created with a "SelectCast" method & "Print MovieDetails" method. Call SelectCast with the Actors Array then call PrintMovieDetails. Once all classes & objects are created run code on the provided Actors.txt file to achieve desired output.

Step 2: Understandings

What do I know?

- I know how to create classes & objects
- I know how to create Polymorphic Arrays
- I know how to loop through Arrays
- I know how to call methods, etc.
- I know how to format text

Step 2 (cont.)

- I know how to display info/data
- I know how to read from files

What I don't know / Questions?

- Make sure reading from file
a per guidelines `next()` `nextLine()`
- Verify string manipulation commands
to properly format output
- Make sure using "throws" `IOException`
to simplify code.

Step 3: pseudo code for main

Classes

Actor	Hero	Villain	Monste	Droid
Type	Type	Type	type	Type
name	name	name	name	name
Moto	*moto	*Moto	*moto	*moto

MOVIE
Constructor Data from heroes villains actors
Methods SelectCast PrintMovieDetails

Header

Number of Heroes Num Her
Number of Villains Num Vill

loop

type --- Name

next

Step 3 (cont)

main

// open Actors.txt for reading

Scanner New

read 1st Ln / int to get size of array

set j = int

new ActorArray [j];

Loop through file to Add values to array

for (i < j; i++)

ActorArray[i] = Scanner next string

Close file

system display ActorArray with Header
and overridden notes

~~system display Header for movie~~

~~Call data from movie~~
~~- Number of Heroes: x~~
~~- Number of Villains: x~~

Call select cast Method from Movie
Count # of Heroes, Villains
+ total actors (Do Most or Droids)

Call PrintMovieDetails to
display Num of Heroes
Num of Villains
And Cast

Step 4: Lessons Learned

Loved seeing the polymorphism at work especially liked the results of the overridden methods. Good reminder of formatting the text to get it into nice columns. Having 1st line of data file being different from rest of data being different formats (e.g. Mr. Incredible vs. Superman) Presented a small challenge when reading into Array but resolved by identifying type the reading rest of line. Overall, fun project.